

Hierarchical Clustering Notes



Lecture 21 - Hierarchical Clustering(1)....
Presentation

Make Detailed Notes for Heirarical Clustering

I'll convert the lecture deck into structured notes on "Hierarchical Clustering," expanding the slide bullets with clear definitions, steps, examples, and exam-style points while staying grounded in the material.

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Detailed Notes: Hierarchical Clustering

These notes are based on the uploaded lecture on **Unsupervised Learning and Hierarchical Clustering**.

Lecture 21 - Hierarchical Clust...

1. Unsupervised Learning

Unsupervised learning is a type of machine learning where the algorithm is given data **without labels**.

The goal is to discover hidden patterns, natural groupings, structures, or relationships in the data.

In supervised learning, we have inputs and known outputs.

In unsupervised learning, we only have input data.

Common Unsupervised Learning Approaches

Some important unsupervised learning techniques include:

1. Clustering
2. Local Outlier Factor
3. Expectation-Maximization
4. Principal Component Analysis
5. Independent Component Analysis
6. Non-negative Matrix Factorization
7. Singular Value Decomposition
8. Isolation Forest

The lecture mainly focuses on **clustering**, especially **hierarchical clustering**.

2. Clustering

Definition

Clustering is an unsupervised learning technique used to divide unlabeled data into groups called **clusters**.

The goal is to create clusters such that:

- Data points within the same cluster are **highly similar**.
- Data points in different clusters are **very different**.
- The algorithm discovers natural categories without using predefined labels.

For example, if we have customer shopping data, clustering can group customers into categories such as budget shoppers, premium buyers, frequent buyers, and occasional buyers.

Important Terms in Clustering

Cluster

A **cluster** is a group of data points that are similar to each other.

Similarity

Similarity means how close or related two data points are.

Dissimilarity / Distance

Dissimilarity measures how different two points are.

In clustering, distance metrics are often used to measure dissimilarity.

Common distance measures include:

Distance Measure	Meaning
Euclidean Distance	Straight-line distance between two points
Manhattan Distance	Distance measured along grid-like paths
Cosine Distance	Measures angle-based difference, often used in text data
Jaccard Distance	Used for sets or binary data
Correlation Distance	Measures difference in pattern or trend

3. Review: K-Means Clustering

Before hierarchical clustering, the lecture reviews **K-Means Clustering**.

What is K-Means?

K-Means is a simple and powerful clustering algorithm.

It tries to group similar data points into **K clusters**.

Here, **K** represents the number of clusters.

For example:

- If $K = 2$, the algorithm creates 2 clusters.
- If $K = 3$, the algorithm creates 3 clusters.
- If $K = 5$, the algorithm creates 5 clusters.

Key Point

In K-Means, the value of **K must be decided before running the algorithm**.

This is one limitation of K-Means because sometimes we do not know the correct number of clusters in advance.

Basic Steps of K-Means

1. Choose the number of clusters, K .
 2. Randomly initialize K centroids.
 3. Assign each data point to the nearest centroid.
 4. Recalculate the centroid of each cluster.
 5. Repeat assignment and centroid update steps until clusters become stable.
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Centroid

A **centroid** is the center point of a cluster.

It is usually calculated as the mean of all points in that cluster.

4. Centroid Initialization Problem in K-Means

K-Means is sensitive to the initial selection of centroids.

Since the initial centroids are often chosen randomly, different runs of K-Means can produce different final clusters.

This is called **initialization sensitivity**.

How to Reduce Initialization Sensitivity

The lecture mentions two main methods:

1. Repeat K-Means

Run K-Means multiple times with different random initial centroids.

Then select the clustering result that has:

- Small intra-cluster distance
- Large inter-cluster distance

2. K-Means++

K-Means++ is a smarter method for choosing initial centroids.

It tries to select initial centroids that are far apart from each other, which improves the quality and stability of clustering.

K-Means++ Steps

1. Pick the first centroid randomly.
2. Compute the distance of each data point from the selected centroid.
3. Select the next centroid from points that are far from the already selected centroid.
4. Repeat until K centroids are selected.
5. Run the normal K-Means algorithm using these selected centroids.

The main idea is that better starting centroids usually lead to better final clusters.

5. Hierarchical Clustering

Definition

Hierarchical clustering is a clustering method that builds a hierarchy of clusters.

Instead of directly dividing data into a fixed number of clusters, hierarchical clustering creates a tree-like structure showing how clusters are related to each other.

This tree-like structure is called a **dendrogram**.

Main Idea

Hierarchical clustering organizes data at multiple levels.

At one level, every point may be its own cluster.

At another level, similar points are merged into larger clusters.

At the highest level, all points may belong to one large cluster.

This makes hierarchical clustering useful when we want to understand the structure of data at different levels of detail.

Example: Biological Taxonomy

The lecture gives an example similar to biological classification.

For example:

```
Animal
├── Vertebrate
│   ├── Fish
│   ├── Reptile
│   ├── Amphibian
│   └── Mammal
└── Invertebrate
    └── Worm
```



This is hierarchical because the broader category **Animal** is divided into smaller categories such as **Vertebrate** and **Invertebrate**, which are further divided into more specific groups.

Similarly, hierarchical clustering divides or combines data points to form a tree of clusters.

6. Types of Hierarchical Clustering

There are two main types:

1. Divisive Clustering
 2. Agglomerative Clustering
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7. Divisive Clustering

Definition

Divisive clustering is a **top-down** hierarchical clustering method.

It starts with all data points in one large cluster and then repeatedly splits clusters into smaller clusters.

Divisive Clustering Process

The process begins with one cluster containing all data points.

Then the algorithm splits this large cluster into smaller clusters.

The splitting continues until:

- The desired number of clusters is obtained, or
 - Each data point becomes its own cluster, or
 - A stopping condition is reached.
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Steps of Divisive Clustering

1. Start with all data points in one single cluster.
 2. Find the cluster that should be split.
 3. Divide that cluster into two least similar clusters.
 4. Repeat the splitting process recursively.
 5. Stop when the desired number of clusters is obtained.
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Use of Sum of Squared Errors

The lecture mentions using **Sum of Squared Errors**, or **SSE**, to decide which cluster should be split.

What is SSE?

SSE measures how spread out the points in a cluster are.

A high SSE means the points in that cluster are far from the cluster center.

This suggests that the cluster may contain multiple smaller groups and should be split.

Formula

For a cluster:

$$SSE = \sum_{i=1}^n (x_i - c)^2$$

Where:

- x_i is a data point.
- c is the cluster centroid.
- n is the number of points in the cluster.

For multidimensional data, distance from the centroid is usually computed using Euclidean distance.

Divisive Clustering Example

Suppose we have six data points:

A, B, C, D, E, F



Initially:

{A, B, C, D, E, F}



After first split:

{A, B, C} and {D, E, F}



After next split:

{A, B}, {C}, {D, E, F}



After another split:

{A, B}, {C}, {D, E}, {F}



This continues until the desired number of clusters is reached.

Advantages of Divisive Clustering

- Gives a complete hierarchy from one cluster to many clusters.
- Useful when the global structure of the dataset is important.
- Can sometimes produce meaningful top-level separation.

Disadvantages of Divisive Clustering

- Computationally expensive.
 - Harder to implement than agglomerative clustering.
 - Splitting decisions are difficult.
 - Once a split is made, it usually cannot be undone.
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8. Agglomerative Clustering

Definition

Agglomerative clustering is a **bottom-up** hierarchical clustering method.

It starts with each data point as its own individual cluster.

Then it repeatedly merges the closest clusters until the desired number of clusters is obtained.

This is the most commonly used type of hierarchical clustering.

Agglomerative Clustering Process

At the beginning:

Each data point = one cluster



Then the algorithm finds the two closest clusters and merges them.

This process continues until:

- All points are in one cluster, or
 - The desired number of clusters is reached.
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Steps of Agglomerative Clustering

1. Start with every data point as a separate cluster.
 2. Calculate the distance between all clusters.
 3. Find the two nearest clusters.
 4. Merge those two clusters into a single cluster.
 5. Update the distance matrix.
 6. Repeat the process until the desired number of clusters is obtained.
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Agglomerative Clustering Example

Suppose we have five points:

A, B, C, D, E



Initially:

{A}, {B}, {C}, {D}, {E}



If A and B are closest, merge them:

{A, B}, {C}, {D}, {E}



If D and E are closest, merge them:

{A, B}, {C}, {D, E}



If C is closest to {D, E}, merge them:

{A, B}, {C, D, E}



Finally:

{A, B, C, D, E}



The complete merging history can be shown using a dendrogram.

9. Agglomerative vs Divisive Clustering

Feature	Agglomerative Clustering	Divisive Clustering
Direction	Bottom-up	Top-down
Starting point	Each point is its own cluster	All points are in one cluster
Main operation	Merge closest clusters	Split least similar clusters
Ending point	One big cluster or desired number of clusters	Individual clusters or desired number of clusters
Common usage	More common	Less common
Complexity	Usually simpler	Usually more expensive
Example operation	Combine A and B	Split large group into two

10. Dendrogram

Definition

A **dendrogram** is a tree-like diagram that shows the hierarchy of clusters.

It visually represents how clusters are merged or split during hierarchical clustering.

Structure of a Dendrogram

A dendrogram usually has:

- Data points at the bottom.
 - Merges or splits shown as branches.
 - Height representing distance or dissimilarity.
 - Larger heights meaning clusters are less similar.
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How to Read a Dendrogram

In agglomerative clustering:

- Leaves represent individual data points.
 - Branches show cluster merging.
 - The height of a merge shows how far apart the merged clusters were.
 - A horizontal cut through the dendrogram gives the final clusters.
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Cutting a Dendrogram

To get a specific number of clusters, we cut the dendrogram at a chosen height.

For example:

- A high cut gives fewer clusters.
- A low cut gives more clusters.

Example

If a horizontal line cuts the dendrogram into three separate branches, then the result is three clusters.

11. Linkage Criteria

In hierarchical clustering, we need to decide how to measure the distance between two clusters.

This is called the **linkage criterion**.

Different linkage methods can produce different clustering results.

1. Single Linkage

Single linkage defines the distance between two clusters as the distance between their closest points.

$$D(A, B) = \min d(a, b)$$

Where:

- A and B are clusters.
- a is a point in cluster A .
- b is a point in cluster B .

Advantage

Can detect long, chain-like clusters.

Disadvantage

Sensitive to noise and outliers.

2. Complete Linkage

Complete linkage defines the distance between two clusters as the distance between their farthest points.

$$D(A, B) = \max d(a, b)$$

Advantage

Produces compact clusters.

Disadvantage

Can break large natural clusters if some points are far apart.

3. Average Linkage

Average linkage defines the distance between two clusters as the average distance between all pairs of points in the two clusters.

$$D(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$

Advantage

More balanced than single and complete linkage.

Disadvantage

More computationally expensive.

4. Centroid Linkage

Centroid linkage defines the distance between two clusters as the distance between their centroids.

$$D(A, B) = d(c_A, c_B)$$

Where:

- c_A is the centroid of cluster A.
- c_B is the centroid of cluster B.

Advantage

Simple and intuitive.

Disadvantage

Can sometimes produce confusing dendrograms because cluster distances may decrease after merging.

5. Ward's Linkage

Ward's linkage merges clusters in a way that minimizes the increase in total within-cluster variance.

It is related to SSE.

Advantage

Usually creates compact and well-separated clusters.

Disadvantage

Works best with Euclidean distance and numerical data.

12. Distance Matrix

A **distance matrix** stores the pairwise distances between all data points or clusters.

For example, for four points A, B, C, and D:

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

The diagonal values are zero because the distance of a point from itself is zero.

In agglomerative clustering, the smallest non-zero value in the distance matrix tells us which two clusters should be merged first.

In the example above:

- A and B have distance 2.
- C and D have distance 4.

So A and B would be merged first.

13. Stopping Conditions in Hierarchical Clustering

Hierarchical clustering can stop based on different conditions.

Common stopping conditions include:

1. Stop when the desired number of clusters is reached.
 2. Stop when the distance between clusters becomes too large.
 3. Stop when each point is in its own cluster.
 4. Stop when all points are merged into one cluster.
 5. Stop using a dendrogram cut.
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14. Choosing the Number of Clusters

Hierarchical clustering does not always require the number of clusters in advance.

Instead, we can create a dendrogram first and then decide the number of clusters by cutting the tree.

Methods for Choosing Clusters

1. Visual Inspection of Dendrogram

Look for large vertical gaps in the dendrogram.

A large gap means that two clusters being merged are quite different.

Cutting before that large jump often gives a good number of clusters.

2. Domain Knowledge

Sometimes the correct number of clusters depends on the application.

For example, in customer segmentation, a business may want exactly 3 or 4 customer groups.

3. Evaluation Metrics

Possible clustering evaluation metrics include:

- Silhouette score
 - Davies-Bouldin index
 - Calinski-Harabasz index
 - Intra-cluster distance
 - Inter-cluster distance
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15. Intra-cluster and Inter-cluster Distance

Intra-cluster Distance

This measures how close points are within the same cluster.

Good clustering should have **low intra-cluster distance**.

Inter-cluster Distance

This measures how far different clusters are from each other.

Good clustering should have **high inter-cluster distance**.

So, a good clustering solution has:

Low intra-cluster distance

High inter-cluster distance



16. Comparison: K-Means vs Hierarchical Clustering

Feature	K-Means Clustering	Hierarchical Clustering
Type	Partitional clustering	Hierarchical clustering
Number of clusters	K must be given before running	Can be decided later using dendrogram
Output	Flat clusters	Tree of clusters
Cluster shape	Works best with spherical clusters	Depends on linkage method
Initialization	Sensitive to initial centroids	No random centroid initialization required
Interpretability	Less visual	More interpretable using dendrogram
Scalability	Faster for large datasets	Slower for large datasets
Example result	3 separate clusters	Full hierarchy of clusters

17. Advantages of Hierarchical Clustering

Hierarchical clustering has several benefits:

1. It does not always require specifying the number of clusters in advance.
2. It produces a dendrogram, which is easy to visualize.
3. It shows relationships between clusters at different levels.
4. It can be used with different distance metrics.
5. It is useful for small to medium-sized datasets.
6. It gives more information than simple flat clustering.

18. Disadvantages of Hierarchical Clustering

Some limitations are:

1. It can be computationally expensive for large datasets.
2. Once a merge or split is made, it usually cannot be undone.
3. It is sensitive to noise and outliers.
4. Different linkage methods can produce different results.

- 5. Choosing the correct cut point in a dendrogram can be subjective.
- 6. It may not work well when clusters overlap heavily.

19. Applications of Hierarchical Clustering

Hierarchical clustering is used in many areas.

1. Biology

Used for taxonomy, gene expression analysis, and grouping species.

2. Marketing

Used for customer segmentation.

3. Document Clustering

Used to group similar documents, articles, or research papers.

4. Image Analysis

Used to group similar images or image regions.

5. Social Network Analysis

Used to detect communities or groups in social networks.

6. Recommendation Systems

Used to group similar users or items.

7. Anomaly Detection

Unusual points may appear far away from other clusters.

20. Simple Numerical Example

Suppose we have four one-dimensional data points:

A = 1
B = 2
C = 8
D = 10



Using Euclidean distance:

Pair	Distance
A-B	1
A-C	7
A-D	9
B-C	6

Pair	Distance
B-D	8
C-D	2

Step 1

Each point starts as its own cluster:

{A}, {B}, {C}, {D}



Step 2

The closest pair is A and B because their distance is 1.

Merge A and B:

{A, B}, {C}, {D}



Step 3

The next closest pair is C and D because their distance is 2.

Merge C and D:

{A, B}, {C, D}



Step 4

Now we have two clusters:

{A, B} and {C, D}



If we want two clusters, we stop here.

If we continue, both clusters are eventually merged into one cluster:

{A, B, C, D}



This process can be represented using a dendrogram.

21. Key Exam Points

Remember these important points:

1. **Hierarchical clustering** creates a tree-like hierarchy of clusters.
2. The tree diagram is called a **dendrogram**.
3. There are two main types: **agglomerative** and **divisive**.
4. Agglomerative clustering is **bottom-up**.
5. Divisive clustering is **top-down**.
6. Agglomerative starts with each point as its own cluster.

7. Divisive starts with all points in one cluster.
 8. In agglomerative clustering, the two closest clusters are merged.
 9. In divisive clustering, clusters are recursively split.
 10. A dendrogram can be cut at a certain height to obtain the desired number of clusters.
 11. Hierarchical clustering does not always require choosing K in advance.
 12. K-Means requires K before running the algorithm.
 13. The choice of distance metric and linkage method strongly affects results.
 14. Good clustering has low intra-cluster distance and high inter-cluster distance.
 15. SSE can be used to decide which cluster to split in divisive clustering.
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22. Common Short Questions and Answers

Q1. What is hierarchical clustering?

Hierarchical clustering is a clustering technique that builds a tree-like structure of clusters, showing how data points or groups are related at different levels.

Q2. What is a dendrogram?

A dendrogram is a tree diagram that represents the hierarchy of clusters formed during hierarchical clustering.

Q3. What is agglomerative clustering?

Agglomerative clustering is a bottom-up method where each data point starts as its own cluster, and the closest clusters are merged step by step.

Q4. What is divisive clustering?

Divisive clustering is a top-down method where all data points start in one cluster, and the cluster is repeatedly split into smaller clusters.

Q5. What is the main difference between K-Means and hierarchical clustering?

K-Means requires the number of clusters K before running. Hierarchical clustering creates a dendrogram, and the number of clusters can be selected later by cutting the dendrogram.

Q6. Why is K-Means sensitive to initialization?

K-Means starts with randomly selected centroids. Different initial centroids may lead to different final clusters.

Q7. How does K-Means++ help?

K-Means++ chooses initial centroids more carefully, usually far apart from one another, which improves clustering quality.

Q8. What is SSE?

SSE, or Sum of Squared Errors, measures how far data points are from their cluster centroid. A larger SSE means the cluster is more spread out.

23. Final Summary

Hierarchical clustering is an unsupervised learning technique used to group similar data points into a hierarchy of clusters. It is especially useful when we want to understand data at multiple levels rather than only producing one flat clustering result.

There are two main approaches:

Agglomerative clustering = Bottom-up merging
Divisive clustering = Top-down splitting



The most important output of hierarchical clustering is the **dendrogram**, which helps visualize how clusters are formed and how many clusters may be appropriate.

Compared with K-Means, hierarchical clustering is more interpretable but usually slower for large datasets. It is widely used in fields such as biology, marketing, document analysis, image processing, and social network analysis.

        Sources